

Held-out test results

Generated by scripts/make_results_md.py from results/test. 400 test chips, Paris and Khartoum, frozen settings from configs/frozen.yaml. F1 in percent, 95% block-bootstrap intervals.

F1 vs resolution, IoU 0.25, area

GSD (m)	OWL-ViT F1 [95% CI]	OWLv2 F1 [95% CI]	Colour rule F1 [95% CI]	OWL-ViT chance	OWLv2 chance	Colour chance
0.3	36.5 [34.0, 38.7]	32.7 [30.0, 34.9]	3.7 [2.6, 4.5]	11.1	8.7	0.3
0.45	38.0 [35.6, 40.1]	35.9 [33.3, 38.1]	5.9 [3.8, 7.8]	11.5	9.7	0.6
0.6	38.5 [36.0, 40.9]	37.4 [34.3, 39.9]	6.0 [3.9, 7.9]	11.6	9.8	0.6
0.9	39.9 [36.9, 42.4]	39.0 [35.0, 42.2]	6.4 [4.1, 8.5]	11.6	9.3	0.6
1.2	39.8 [36.6, 42.5]	37.8 [33.2, 41.6]	6.3 [4.0, 8.5]	11.3	8.1	0.6
1.8	33.8 [29.3, 37.3]	33.7 [27.9, 38.5]	5.9 [3.7, 8.0]	8.3	7.0	0.5
2.4	16.5 [13.1, 19.5]	28.6 [21.3, 34.9]	5.6 [3.5, 7.6]	3.4	5.3	0.5
3.6	1.0 [0.6, 1.5]	5.0 [2.8, 7.2]	4.4 [2.7, 6.0]	0.2	0.9	0.5
5	0.0 [0.0, 0.1]	0.2 [0.1, 0.4]	3.0 [1.9, 4.1]	0.0	0.0	0.3
10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.4 [0.2, 0.6]	0.0	0.0	0.1

F1 vs resolution, IoU 0.25, blur_resize

GSD (m)	OWL-ViT F1 [95% CI]	OWLv2 F1 [95% CI]	Colour rule F1 [95% CI]	OWL-ViT chance	OWLv2 chance	Colour chance
0.3	36.5 [34.0, 38.7]	32.7 [30.0, 34.9]	3.7 [2.6, 4.5]	11.1	8.7	0.3
0.45	39.7 [37.2, 41.9]	38.4 [35.2, 40.9]	6.2 [4.0, 8.3]	11.6	9.9	0.6
0.6	40.6 [37.7, 43.0]	39.5 [35.5, 42.4]	6.3 [4.0, 8.5]	11.5	9.5	0.6
0.9	40.6 [37.4, 43.2]	38.4 [33.9, 41.8]	6.2 [3.9, 8.3]	11.3	8.7	0.6
1.2	39.2 [35.1, 42.5]	34.1 [28.6, 38.3]	5.7 [3.6, 7.7]	10.3	7.3	0.6
1.8	24.5 [19.8, 28.3]	20.5 [15.1, 25.5]	5.2 [3.2, 7.2]	5.2	3.7	0.5
2.4	5.2 [3.6, 6.8]	9.2 [5.9, 12.5]	4.5 [2.8, 6.2]	1.0	1.4	0.5
3.6	0.2 [0.1, 0.4]	0.1 [0.0, 0.2]	2.9 [1.9, 4.0]	0.0	0.0	0.3
5	0.0 [0.0, 0.1]	0.0 [0.0, 0.0]	1.3 [0.8, 1.8]	0.0	0.0	0.2
10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	0.0	0.0	0.0

F1 vs resolution, IoU 0.5, area

GSD (m)	OWL-ViT F1 [95% CI]	OWLv2 F1 [95% CI]	Colour rule F1 [95% CI]	OWL-ViT chance	OWLv2 chance	Colour chance
0.3	17.2 [15.7, 18.8]	16.5 [15.3, 17.6]	1.1 [0.7, 1.5]	1.5	1.0	0.0
0.45	17.5 [16.0, 19.1]	15.5 [14.0, 17.2]	1.9 [1.2, 2.6]	1.4	1.1	0.1
0.6	17.7 [16.1, 19.4]	16.5 [14.9, 18.4]	2.0 [1.2, 2.8]	1.4	1.0	0.1
0.9	17.9 [16.3, 19.5]	15.0 [13.4, 16.9]	2.0 [1.2, 2.7]	1.4	1.0	0.1
1.2	17.3 [15.5, 19.0]	13.9 [12.1, 15.7]	1.9 [1.2, 2.6]	1.3	0.8	0.1
1.8	13.6 [11.6, 15.6]	11.4 [9.5, 13.2]	1.6 [1.0, 2.2]	0.9	0.7	0.1
2.4	6.3 [4.9, 7.6]	9.1 [6.9, 11.0]	1.4 [1.0, 1.9]	0.3	0.6	0.1
3.6	0.4 [0.2, 0.7]	1.7 [1.0, 2.5]	1.1 [0.7, 1.5]	0.0	0.1	0.0
5	0.0 [0.0, 0.1]	0.1 [0.0, 0.2]	0.7 [0.4, 1.1]	0.0	0.0	0.0
10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	0.0	0.0	0.0

F1 vs resolution, IoU 0.5, blur_resize

GSD (m)	OWL-ViT F1 [95% CI]	OWLv2 F1 [95% CI]	Colour rule F1 [95% CI]	OWL-ViT chance	OWLv2 chance	Colour chance
0.3	17.2 [15.7, 18.8]	16.5 [15.3, 17.6]	1.1 [0.7, 1.5]	1.5	1.0	0.0
0.45	18.1 [16.5, 19.8]	16.0 [14.4, 17.8]	1.9 [1.2, 2.7]	1.5	1.1	0.1
0.6	18.2 [16.6, 19.9]	15.6 [13.9, 17.6]	1.8 [1.1, 2.5]	1.4	1.0	0.1
0.9	18.1 [16.3, 20.0]	14.2 [12.5, 16.2]	1.7 [1.1, 2.3]	1.3	0.9	0.1
1.2	16.6 [14.6, 18.7]	11.7 [9.8, 13.6]	1.6 [1.0, 2.3]	1.2	0.7	0.1
1.8	9.6 [7.8, 11.4]	7.1 [5.2, 8.9]	1.3 [0.8, 1.7]	0.5	0.3	0.0
2.4	1.9 [1.3, 2.7]	3.4 [2.1, 4.6]	1.0 [0.7, 1.4]	0.1	0.1	0.0
3.6	0.1 [0.0, 0.2]	0.1 [0.0, 0.2]	0.7 [0.5, 1.0]	0.0	0.0	0.0
5	0.0 [0.0, 0.1]	0.0 [0.0, 0.0]	0.5 [0.2, 0.7]	0.0	0.0	0.0
10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.1 [0.0, 0.2]	0.0	0.0	0.0

Break points (F1 falls to half of its peak above chance)

Detector	Method	IoU	Setting	Peak GSD	Peak F1	Break GSD [95% CI]	Break vs finest level
OWL-ViT	area	0.25	fixed	0.9	39.9	2.34 [2.19, 2.47]	2.43
OWL-ViT	area	0.25	per_level	0.45	43.5	3.46 [3.04, 3.77]	3.53
OWL-ViT	area	0.5	fixed	0.9	17.9	2.18 [2.06, 2.28]	2.21
OWL-ViT	area	0.5	per_level	0.3	20.2	2.51 [2.29, 2.72]	2.51
OWL-ViT	blur_resize	0.25	fixed	0.6	40.6	1.97 [1.85, 2.03]	2.04
OWL-ViT	blur_resize	0.25	per_level	0.45	43.5	2.42 [2.23, 2.75]	2.49
OWL-ViT	blur_resize	0.5	fixed	0.6	18.2	1.85 [1.71, 1.93]	1.89
OWL-ViT	blur_resize	0.5	per_level	0.45	20.2	2.09 [1.98, 2.18]	2.09
OWLv2	area	0.25	fixed	0.9	39.0	2.87 [2.59, 3.02]	3.05
OWLv2	area	0.25	per_level	0.6	39.0	7.24 [6.51, 8.19]	7.56
OWLv2	area	0.5	fixed	0.6	16.5	2.52 [2.05, 2.73]	2.52
OWLv2	area	0.5	per_level	0.3	17.6	4.32 [3.82, 4.82]	4.32
OWLv2	blur_resize	0.25	fixed	0.6	39.5	1.90 [1.64, 2.10]	2.09
OWLv2	blur_resize	0.25	per_level	0.45	38.4	5.15 [2.26, 5.84]	5.29
OWLv2	blur_resize	0.5	fixed	0.3	16.5	1.64 [1.41, 1.92]	1.64
OWLv2	blur_resize	0.5	per_level	0.3	17.6	2.16 [1.67, 2.78]	2.16
Colour rule	area	0.25	fixed	0.9	6.4	4.77 [4.13, 5.63]	6.76
Colour rule	area	0.25	per_level	0.9	6.4	4.77 [4.13, 5.63]	6.76
Colour rule	area	0.5	fixed	0.6	2.0	3.81 [2.92, 5.43]	6.14
Colour rule	area	0.5	per_level	0.6	2.0	3.81 [2.92, 5.43]	6.14
Colour rule	blur_resize	0.25	fixed	0.6	6.3	3.39 [2.90, 3.92]	4.46
Colour rule	blur_resize	0.25	per_level	0.6	6.3	3.39 [2.90, 3.92]	4.46
Colour rule	blur_resize	0.5	fixed	0.45	1.9	2.63 [1.78, 4.36]	4.45

Detector	Method	IoU	Setting	Peak GSD	Peak F1	Break GSD [95% CI]	Break vs finest level
Colour rule	blur_resize	0.5	per_level	0.45	1.9	2.63 [1.75, 4.36]	4.45

Native 1.2 m multispectral vs simulated 1.2 m (same 400 chips)

Detector	IoU	Simulation	Native F1	Simulated F1	Native minus simulated [95% CI]
OWL-ViT	0.25	area	40.8	39.8	1.0 [-0.2, 2.0]
OWL-ViT	0.25	blur_resize	40.8	39.2	1.5 [0.1, 3.1]
OWL-ViT	0.5	area	16.9	17.3	-0.4 [-1.3, 0.6]
OWL-ViT	0.5	blur_resize	16.9	16.6	0.3 [-0.9, 1.4]
OWLv2	0.25	area	41.1	37.8	3.3 [1.6, 5.4]
OWLv2	0.25	blur_resize	41.1	34.1	7.1 [4.8, 10.0]
OWLv2	0.5	area	17.0	13.9	3.1 [1.9, 4.3]
OWLv2	0.5	blur_resize	17.0	11.7	5.3 [4.2, 6.5]
Colour rule	0.25	area	8.7	6.3	2.4 [1.4, 3.2]
Colour rule	0.25	blur_resize	8.7	5.7	2.9 [2.0, 3.8]
Colour rule	0.5	area	3.7	1.9	1.8 [1.1, 2.3]
Colour rule	0.5	blur_resize	3.7	1.6	2.0 [1.2, 2.7]

Detector F1 against the oracle-location baseline, IoU 0.25, area

The oracle keeps each detector's box count and sizes and moves the boxes onto random true building centres. Difference is detector minus oracle, with a block-bootstrap interval.

Detector	Setting	GSD (m)	F1	Oracle F1	Difference [95% CI]
Colour rule	fixed	0.3	3.7	4.1	-0.5 [-0.7, -0.2]
Colour rule	fixed	0.45	5.9	6.8	-0.9 [-1.4, -0.3]
Colour rule	fixed	0.6	6.0	6.9	-0.9 [-1.5, -0.4]
Colour rule	fixed	0.9	6.4	7.6	-1.2 [-1.9, -0.5]
Colour rule	fixed	1.2	6.3	7.2	-0.9 [-1.5, -0.3]
Colour rule	fixed	1.8	5.9	6.9	-1.0 [-1.7, -0.4]
Colour rule	fixed	2.4	5.6	6.3	-0.8 [-1.2, -0.3]
Colour rule	fixed	3.6	4.4	5.1	-0.7 [-1.2, -0.3]
Colour rule	fixed	5	3.0	3.7	-0.7 [-1.2, -0.2]
Colour rule	fixed	10	0.4	0.6	-0.3 [-0.5, -0.1]
Colour rule	per_level	0.3	3.7	4.1	-0.5 [-0.7, -0.2]
Colour rule	per_level	0.45	5.9	6.8	-0.9 [-1.4, -0.3]
Colour rule	per_level	0.6	6.0	6.9	-0.9 [-1.5, -0.4]
Colour rule	per_level	0.9	6.4	7.6	-1.2 [-1.9, -0.5]
Colour rule	per_level	1.2	6.3	7.2	-0.9 [-1.5, -0.3]
Colour rule	per_level	1.8	5.8	6.9	-1.1 [-1.7, -0.5]
Colour rule	per_level	2.4	5.6	6.3	-0.8 [-1.2, -0.3]
Colour rule	per_level	3.6	4.4	5.1	-0.7 [-1.2, -0.3]
Colour rule	per_level	5	3.0	3.7	-0.7 [-1.2, -0.2]
Colour rule	per_level	10	0.4	0.6	-0.3 [-0.5, -0.1]
OWLv2	fixed	0.3	32.7	36.3	-3.5 [-4.3, -2.8]
OWLv2	fixed	0.45	35.9	38.7	-2.7 [-3.6, -1.7]
OWLv2	fixed	0.6	37.4	39.7	-2.4 [-3.3, -1.3]
OWLv2	fixed	0.9	39.0	40.7	-1.7 [-3.0, -0.2]
OWLv2	fixed	1.2	37.8	38.9	-1.1 [-2.6, 0.5]
OWLv2	fixed	1.8	33.7	33.8	-0.1 [-1.3, 1.2]
OWLv2	fixed	2.4	28.6	26.9	1.7 [0.4, 2.8]
OWLv2	fixed	3.6	5.0	4.0	1.0 [0.4, 1.6]
OWLv2	fixed	5	0.2	0.1	0.1 [0.0, 0.1]
OWLv2	fixed	10	0.0	0.0	0.0 [0.0, 0.0]
OWLv2	per_level	0.3	37.8	41.0	-3.3 [-4.5, -1.9]
OWLv2	per_level	0.45	37.9	39.6	-1.7 [-3.1, -0.0]
OWLv2	per_level	0.6	39.0	40.3	-1.3 [-3.0, 0.6]
OWLv2	per_level	0.9	38.6	39.9	-1.3 [-2.8, 0.5]
OWLv2	per_level	1.2	37.8	38.9	-1.1 [-2.7, 0.6]
OWLv2	per_level	1.8	37.4	42.1	-4.6 [-5.9, -3.1]
OWLv2	per_level	2.4	36.4	41.4	-5.1 [-6.4, -3.5]
OWLv2	per_level	3.6	32.4	39.2	-6.8 [-8.3, -5.4]
OWLv2	per_level	5	27.3	34.6	-7.3 [-9.3, -5.3]
OWLv2	per_level	10	16.3	18.1	-1.8 [-3.0, -0.6]
OWL-ViT	fixed	0.3	36.5	41.4	-4.9 [-5.6, -4.1]

Detector	Setting	GSD (m)	F1	Oracle F1	Difference [95% CI]
OWL-ViT	fixed	0.45	38.0	42.7	-4.7 [-5.5, -3.8]
OWL-ViT	fixed	0.6	38.5	43.5	-5.0 [-5.8, -4.2]
OWL-ViT	fixed	0.9	39.9	45.2	-5.3 [-6.4, -4.2]
OWL-ViT	fixed	1.2	39.8	46.8	-7.0 [-8.4, -5.5]
OWL-ViT	fixed	1.8	33.8	36.9	-3.1 [-4.5, -1.6]
OWL-ViT	fixed	2.4	16.5	14.9	1.6 [0.8, 2.4]
OWL-ViT	fixed	3.6	1.0	0.9	0.1 [-0.1, 0.4]
OWL-ViT	fixed	5	0.0	0.0	0.0 [-0.0, 0.0]
OWL-ViT	fixed	10	0.0	0.0	0.0 [0.0, 0.0]
OWL-ViT	per_level	0.3	43.0	50.2	-7.2 [-8.5, -5.7]
OWL-ViT	per_level	0.45	43.5	49.9	-6.4 [-7.7, -4.9]
OWL-ViT	per_level	0.6	43.4	48.9	-5.5 [-6.9, -3.8]
OWL-ViT	per_level	0.9	41.8	46.1	-4.4 [-6.0, -2.7]
OWL-ViT	per_level	1.2	39.8	46.8	-7.0 [-8.3, -5.3]
OWL-ViT	per_level	1.8	34.6	41.8	-7.2 [-8.2, -6.2]
OWL-ViT	per_level	2.4	30.7	31.3	-0.6 [-1.7, 0.6]
OWL-ViT	per_level	3.6	22.0	22.8	-0.8 [-1.8, 0.2]
OWL-ViT	per_level	5	10.2	8.3	1.9 [1.2, 2.7]
OWL-ViT	per_level	10	2.8	1.7	1.1 [0.7, 1.6]

Detector F1 against the oracle-location baseline, IoU 0.25, blur_resize

The oracle keeps each detector's box count and sizes and moves the boxes onto random true building centres. Difference is detector minus oracle, with a block-bootstrap interval.

Detector	Setting	GSD (m)	F1	Oracle F1	Difference [95% CI]
Colour rule	fixed	0.3	3.7	4.1	-0.5 [-0.7, -0.2]
Colour rule	fixed	0.45	6.2	7.1	-0.9 [-1.5, -0.3]
Colour rule	fixed	0.6	6.3	7.2	-0.9 [-1.6, -0.3]
Colour rule	fixed	0.9	6.2	7.2	-1.1 [-1.6, -0.5]
Colour rule	fixed	1.2	5.7	6.9	-1.1 [-1.7, -0.5]
Colour rule	fixed	1.8	5.2	6.1	-0.9 [-1.4, -0.3]
Colour rule	fixed	2.4	4.5	5.3	-0.7 [-1.3, -0.2]
Colour rule	fixed	3.6	2.9	3.7	-0.7 [-1.3, -0.2]
Colour rule	fixed	5	1.3	1.8	-0.5 [-0.8, -0.2]
Colour rule	fixed	10	0.2	0.2	-0.1 [-0.3, 0.1]
Colour rule	per_level	0.3	3.7	4.1	-0.5 [-0.7, -0.2]
Colour rule	per_level	0.45	6.2	7.1	-0.9 [-1.5, -0.3]
Colour rule	per_level	0.6	6.3	7.2	-0.9 [-1.5, -0.3]
Colour rule	per_level	0.9	6.2	7.2	-1.1 [-1.6, -0.5]
Colour rule	per_level	1.2	5.7	6.9	-1.1 [-1.8, -0.5]
Colour rule	per_level	1.8	5.1	6.1	-0.9 [-1.4, -0.4]
Colour rule	per_level	2.4	4.5	5.3	-0.7 [-1.2, -0.2]
Colour rule	per_level	3.6	2.9	3.7	-0.7 [-1.2, -0.2]
Colour rule	per_level	5	1.3	1.8	-0.5 [-0.8, -0.3]
Colour rule	per_level	10	0.2	0.2	-0.1 [-0.3, 0.1]
OWLv2	fixed	0.3	32.7	36.3	-3.5 [-4.3, -2.8]
OWLv2	fixed	0.45	38.4	40.5	-2.1 [-3.2, -1.0]
OWLv2	fixed	0.6	39.5	41.7	-2.2 [-3.6, -0.8]
OWLv2	fixed	0.9	38.4	39.8	-1.5 [-3.0, 0.1]
OWLv2	fixed	1.2	34.1	34.9	-0.8 [-2.1, 0.7]
OWLv2	fixed	1.8	20.5	17.7	2.8 [1.4, 4.0]
OWLv2	fixed	2.4	9.2	6.9	2.2 [1.3, 3.2]
OWLv2	fixed	3.6	0.1	0.1	0.0 [-0.0, 0.1]
OWLv2	fixed	5	0.0	0.0	0.0 [0.0, 0.0]
OWLv2	fixed	10	0.0	0.0	0.0 [0.0, 0.0]
OWLv2	per_level	0.3	37.8	41.0	-3.3 [-4.5, -1.9]
OWLv2	per_level	0.45	38.4	38.7	-0.4 [-1.7, 1.2]
OWLv2	per_level	0.6	37.0	35.4	1.6 [0.3, 3.0]
OWLv2	per_level	0.9	37.2	38.2	-0.9 [-2.5, 0.8]
OWLv2	per_level	1.2	34.1	34.9	-0.8 [-2.2, 0.6]
OWLv2	per_level	1.8	30.5	31.6	-1.1 [-2.3, 0.1]
OWLv2	per_level	2.4	23.1	21.9	1.2 [0.1, 2.2]
OWLv2	per_level	3.6	19.7	21.5	-1.9 [-3.6, -0.2]
OWLv2	per_level	5	20.1	25.0	-4.9 [-6.8, -3.2]
OWLv2	per_level	10	7.2	5.8	1.3 [0.5, 2.1]
OWL-ViT	fixed	0.3	36.5	41.4	-4.9 [-5.6, -4.1]

Detector	Setting	GSD (m)	F1	Oracle F1	Difference [95% CI]
OWL-ViT	fixed	0.45	39.7	44.4	-4.6 [-5.4, -3.8]
OWL-ViT	fixed	0.6	40.6	45.9	-5.3 [-6.4, -4.2]
OWL-ViT	fixed	0.9	40.6	47.1	-6.5 [-7.8, -4.9]
OWL-ViT	fixed	1.2	39.2	45.3	-6.0 [-7.5, -4.4]
OWL-ViT	fixed	1.8	24.5	22.7	1.8 [0.7, 2.7]
OWL-ViT	fixed	2.4	5.2	4.5	0.7 [0.2, 1.2]
OWL-ViT	fixed	3.6	0.2	0.1	0.1 [0.0, 0.2]
OWL-ViT	fixed	5	0.0	0.0	0.0 [0.0, 0.0]
OWL-ViT	fixed	10	0.0	0.0	0.0 [0.0, 0.0]
OWL-ViT	per_level	0.3	43.0	50.2	-7.2 [-8.5, -5.7]
OWL-ViT	per_level	0.45	43.5	49.3	-5.7 [-7.3, -4.1]
OWL-ViT	per_level	0.6	42.6	46.7	-4.1 [-5.7, -2.4]
OWL-ViT	per_level	0.9	39.3	40.9	-1.6 [-3.1, -0.1]
OWL-ViT	per_level	1.2	39.2	45.3	-6.0 [-7.5, -4.3]
OWL-ViT	per_level	1.8	34.8	41.0	-6.2 [-7.6, -4.7]
OWL-ViT	per_level	2.4	22.6	21.6	1.1 [0.2, 2.0]
OWL-ViT	per_level	3.6	14.0	11.7	2.3 [1.5, 3.1]
OWL-ViT	per_level	5	5.7	4.0	1.7 [1.0, 2.5]
OWL-ViT	per_level	10	1.7	1.0	0.7 [0.3, 1.2]

Detector F1 against the oracle-location baseline, IoU 0.5, area

The oracle keeps each detector's box count and sizes and moves the boxes onto random true building centres. Difference is detector minus oracle, with a block-bootstrap interval.

Detector	Setting	GSD (m)	F1	Oracle F1	Difference [95% CI]
Colour rule	fixed	0.3	1.1	1.7	-0.5 [-0.8, -0.3]
Colour rule	fixed	0.45	1.9	2.8	-0.8 [-1.3, -0.4]
Colour rule	fixed	0.6	2.0	2.8	-0.7 [-1.2, -0.2]
Colour rule	fixed	0.9	2.0	3.0	-1.1 [-1.7, -0.4]
Colour rule	fixed	1.2	1.9	2.9	-1.0 [-1.5, -0.4]
Colour rule	fixed	1.8	1.6	2.7	-1.1 [-1.7, -0.5]
Colour rule	fixed	2.4	1.4	2.5	-1.1 [-1.7, -0.5]
Colour rule	fixed	3.6	1.1	1.8	-0.8 [-1.2, -0.3]
Colour rule	fixed	5	0.7	1.3	-0.5 [-0.9, -0.2]
Colour rule	fixed	10	0.2	0.2	-0.1 [-0.2, 0.1]
Colour rule	per_level	0.3	1.1	1.7	-0.5 [-0.8, -0.3]
Colour rule	per_level	0.45	1.9	2.8	-0.8 [-1.3, -0.4]
Colour rule	per_level	0.6	2.0	2.8	-0.7 [-1.2, -0.2]
Colour rule	per_level	0.9	2.0	3.0	-1.1 [-1.7, -0.4]
Colour rule	per_level	1.2	1.9	2.9	-1.0 [-1.5, -0.4]
Colour rule	per_level	1.8	1.5	2.6	-1.2 [-1.8, -0.5]
Colour rule	per_level	2.4	1.4	2.5	-1.1 [-1.6, -0.5]
Colour rule	per_level	3.6	1.1	1.8	-0.8 [-1.2, -0.3]
Colour rule	per_level	5	0.7	1.3	-0.5 [-0.9, -0.2]
Colour rule	per_level	10	0.2	0.2	-0.1 [-0.2, 0.1]
OWLv2	fixed	0.3	16.5	20.8	-4.4 [-6.2, -2.4]
OWLv2	fixed	0.45	15.5	21.1	-5.6 [-7.7, -2.9]
OWLv2	fixed	0.6	16.5	21.1	-4.6 [-7.1, -1.8]
OWLv2	fixed	0.9	15.0	19.3	-4.3 [-6.5, -1.7]
OWLv2	fixed	1.2	13.9	17.7	-3.8 [-6.0, -1.4]
OWLv2	fixed	1.8	11.4	15.1	-3.6 [-5.3, -1.9]
OWLv2	fixed	2.4	9.1	12.4	-3.3 [-5.0, -1.3]
OWLv2	fixed	3.6	1.7	1.6	0.2 [-0.3, 0.6]
OWLv2	fixed	5	0.1	0.0	0.0 [-0.0, 0.1]
OWLv2	fixed	10	0.0	0.0	0.0 [0.0, 0.0]
OWLv2	per_level	0.3	17.6	20.1	-2.6 [-4.8, -0.0]
OWLv2	per_level	0.45	16.4	18.1	-1.7 [-3.9, 0.6]
OWLv2	per_level	0.6	16.4	18.0	-1.6 [-4.0, 1.0]
OWLv2	per_level	0.9	15.1	18.0	-2.9 [-5.3, -0.2]
OWLv2	per_level	1.2	13.9	17.7	-3.8 [-6.0, -1.3]
OWLv2	per_level	1.8	13.1	20.4	-7.3 [-9.3, -4.7]
OWLv2	per_level	2.4	12.0	20.6	-8.6 [-10.8, -5.9]
OWLv2	per_level	3.6	11.2	20.1	-8.9 [-10.5, -7.1]
OWLv2	per_level	5	7.6	14.9	-7.3 [-8.5, -6.1]
OWLv2	per_level	10	4.3	6.1	-1.7 [-2.6, -1.0]
OWL-ViT	fixed	0.3	17.2	23.9	-6.7 [-8.3, -5.0]

Detector	Setting	GSD (m)	F1	Oracle F1	Difference [95% CI]
OWL-ViT	fixed	0.45	17.5	24.0	-6.5 [-8.0, -4.8]
OWL-ViT	fixed	0.6	17.7	23.8	-6.0 [-7.6, -4.4]
OWL-ViT	fixed	0.9	17.9	23.3	-5.4 [-6.8, -3.9]
OWL-ViT	fixed	1.2	17.3	23.1	-5.8 [-7.2, -4.1]
OWL-ViT	fixed	1.8	13.6	17.0	-3.4 [-5.0, -1.6]
OWL-ViT	fixed	2.4	6.3	6.6	-0.3 [-1.2, 0.5]
OWL-ViT	fixed	3.6	0.4	0.3	0.1 [-0.1, 0.3]
OWL-ViT	fixed	5	0.0	0.0	0.0 [0.0, 0.1]
OWL-ViT	fixed	10	0.0	0.0	0.0 [0.0, 0.0]
OWL-ViT	per_level	0.3	20.2	25.8	-5.6 [-7.5, -3.5]
OWL-ViT	per_level	0.45	20.1	24.8	-4.8 [-6.6, -2.8]
OWL-ViT	per_level	0.6	19.9	23.7	-3.8 [-5.6, -1.8]
OWL-ViT	per_level	0.9	18.9	21.6	-2.7 [-4.1, -1.2]
OWL-ViT	per_level	1.2	17.3	23.1	-5.8 [-7.3, -4.1]
OWL-ViT	per_level	1.8	13.8	22.1	-8.3 [-9.9, -6.4]
OWL-ViT	per_level	2.4	10.7	14.0	-3.3 [-4.9, -1.5]
OWL-ViT	per_level	3.6	6.0	8.8	-2.9 [-4.1, -1.6]
OWL-ViT	per_level	5	3.0	2.4	0.6 [-0.0, 1.2]
OWL-ViT	per_level	10	0.9	0.4	0.5 [0.2, 0.9]

Third city (Las Vegas, 200 chips, never used for tuning), IoU 0.25, area

GSD (m)	OWL-ViT F1 [95% CI]	OWLv2 F1 [95% CI]	Colour rule F1 [95% CI]
0.3	41.5 [39.1, 43.6]	25.6 [24.3, 26.9]	1.4 [0.7, 2.3]
0.45	41.9 [39.5, 44.0]	34.5 [32.7, 36.3]	1.5 [0.7, 2.5]
0.6	42.8 [40.4, 45.0]	36.8 [34.7, 38.8]	1.6 [0.7, 2.7]
0.9	43.7 [41.2, 45.9]	43.0 [40.3, 45.7]	1.6 [0.7, 2.7]
1.2	42.3 [39.5, 44.7]	43.5 [40.8, 46.1]	1.6 [0.7, 2.7]
1.8	33.3 [30.1, 36.1]	42.3 [39.2, 45.2]	1.5 [0.7, 2.6]
2.4	18.8 [15.8, 21.8]	36.7 [32.8, 40.3]	1.5 [0.6, 2.5]
3.6	3.8 [2.7, 5.1]	10.4 [7.0, 14.0]	1.3 [0.5, 2.3]
5	0.8 [0.4, 1.3]	0.7 [0.3, 1.3]	1.1 [0.4, 1.9]
10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.3 [0.0, 0.6]

Third city (Las Vegas), IoU 0.25, blur_resize

GSD (m)	OWL-ViT F1 [95% CI]	OWLv2 F1 [95% CI]	Colour rule F1 [95% CI]
0.3	41.5 [39.1, 43.6]	25.6 [24.3, 26.9]	1.4 [0.7, 2.3]
0.45	43.0 [40.6, 45.1]	38.9 [36.9, 40.8]	1.5 [0.7, 2.6]
0.6	44.2 [41.8, 46.4]	42.8 [40.4, 45.2]	1.6 [0.7, 2.7]
0.9	41.9 [39.0, 44.3]	44.1 [41.3, 46.7]	1.6 [0.7, 2.7]
1.2	37.4 [34.3, 40.0]	44.1 [41.1, 46.9]	1.6 [0.7, 2.7]
1.8	19.4 [16.3, 22.6]	39.1 [35.5, 42.5]	1.4 [0.6, 2.5]
2.4	8.9 [6.9, 11.3]	19.8 [15.6, 24.2]	1.3 [0.5, 2.2]
3.6	0.7 [0.4, 1.2]	1.0 [0.4, 1.8]	1.0 [0.4, 1.7]
5	0.1 [0.0, 0.2]	0.0 [0.0, 0.1]	0.5 [0.1, 0.9]
10	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.1 [0.0, 0.2]

Per-city peak and break point, frozen settings, IoU 0.25

City	Detector	Method	Chips	Peak GSD	Peak F1	Break GSD [95% CI]	F1 at 2.4 m	F1 at 3.6 m
AOI_3_Paris	OWL-ViT	area	220	1.2	45.2	2.56 [2.34, 2.70]	23.4	1.9
AOI_5_Khartoum	OWL-ViT	area	180	0.9	36.8	2.16 [1.95, 2.30]	11.6	0.4
AOI_3_Paris	OWL-ViT	blur_resize	220	1.2	46.8	2.08 [2.00, 2.14]	9.1	0.3
AOI_5_Khartoum	OWL-ViT	blur_resize	180	0.45	37.6	1.78 [1.55, 1.92]	2.6	0.1
AOI_3_Paris	OWLv2	area	220	2.4	46.4	3.14 [3.04, 3.23]	46.4	11.1
AOI_5_Khartoum	OWLv2	area	180	0.6	36.0	2.04 [1.71, 2.28]	12.3	0.8
AOI_3_Paris	OWLv2	blur_resize	220	0.9	45.5	2.32 [2.15, 2.52]	18.7	0.1
AOI_5_Khartoum	OWLv2	blur_resize	180	0.45	34.6	1.42 [1.25, 1.54]	2.3	0.1
AOI_3_Paris	Colour rule	area	220	1.2	13.0	4.75 [4.11, 5.53]	11.7	9.2
AOI_5_Khartoum	Colour rule	area	180	0.9	1.4	6.39 [4.25, 8.44]	1.2	1.0
AOI_3_Paris	Colour rule	blur_resize	220	0.6	12.8	3.37 [2.85, 3.88]	9.6	5.9
AOI_5_Khartoum	Colour rule	blur_resize	180	0.45	1.4	4.33 [2.98, 5.91]	1.0	0.9
AOI_2_Vegas	OWL-ViT	area	200	0.9	43.7	2.31 [2.16, 2.53]	18.8	3.8
AOI_2_Vegas	OWL-ViT	blur_resize	200	0.6	44.2	1.74 [1.60, 1.94]	8.9	0.7
AOI_2_Vegas	OWLv2	area	200	1.2	43.5	3.10 [2.90, 3.32]	36.7	10.4
AOI_2_Vegas	OWLv2	blur_resize	200	1.2	44.1	2.38 [2.21, 2.61]	19.8	1.0
AOI_2_Vegas	Colour rule	area	200	0.6	1.6	6.37 [5.25, 7.13]	1.5	1.3
AOI_2_Vegas	Colour rule	blur_resize	200	0.6	1.6	4.00 [2.95, 4.58]	1.3	1.0

Real Sentinel-2 vs simulated 10 m, Paris and Khartoum (400 chips)

Same chip footprints and labels. Sentinel-2 L2A crops (20 by 20 px), upsampled like the simulation. Oracle F1 is the oracle-location baseline on the real crops.

Detector	Setting	IoU	Simulation	Real S2 F1 [95% CI]	Simulated F1 [95% CI]	Real minus simulated [95% CI]	Oracle F1
OWL-ViT	fixed	0.25	area	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWL-ViT	fixed	0.25	blur_resize	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWL-ViT	fixed	0.5	area	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWL-ViT	fixed	0.5	blur_resize	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWL-ViT	per_level	0.25	area	1.7 [1.2, 2.2]	2.8 [2.0, 3.7]	-1.2 [-1.8, -0.6]	1.2
OWL-ViT	per_level	0.25	blur_resize	1.7 [1.2, 2.2]	1.7 [1.1, 2.4]	-0.0 [-0.5, 0.4]	1.2
OWL-ViT	per_level	0.5	area	0.3 [0.1, 0.6]	0.9 [0.5, 1.3]	-0.5 [-0.8, -0.2]	0.3
OWL-ViT	per_level	0.5	blur_resize	0.3 [0.1, 0.6]	0.7 [0.3, 1.0]	-0.3 [-0.6, -0.0]	0.3
OWLv2	fixed	0.25	area	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWLv2	fixed	0.25	blur_resize	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWLv2	fixed	0.5	area	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWLv2	fixed	0.5	blur_resize	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWLv2	per_level	0.25	area	9.8 [8.1, 11.5]	16.3 [14.5, 18.1]	-6.5 [-8.2, -4.8]	13.8
OWLv2	per_level	0.25	blur_resize	9.8 [8.1, 11.5]	7.2 [5.5, 9.1]	2.6 [1.2, 4.1]	13.8
OWLv2	per_level	0.5	area	1.8 [1.3, 2.3]	4.3 [3.5, 5.2]	-2.6 [-3.3, -1.9]	3.8
OWLv2	per_level	0.5	blur_resize	1.8 [1.3, 2.3]	2.0 [1.3, 2.7]	-0.2 [-0.8, 0.3]	3.8
Colour rule	fixed	0.25	area	1.5 [1.1, 1.8]	0.4 [0.2, 0.6]	1.1 [0.7, 1.5]	2.9
Colour rule	fixed	0.25	blur_resize	1.5 [1.1, 1.8]	0.2 [0.1, 0.3]	1.3 [1.0, 1.7]	2.9
Colour rule	fixed	0.5	area	0.1 [0.1, 0.2]	0.2 [0.1, 0.3]	-0.0 [-0.2, 0.1]	1.0
Colour rule	fixed	0.5	blur_resize	0.1 [0.1, 0.2]	0.1 [0.0, 0.2]	0.0 [-0.1, 0.1]	1.0
Colour rule	per_level	0.25	area	0.7 [0.5, 0.9]	0.4 [0.2, 0.6]	0.3 [-0.0, 0.6]	1.7
Colour rule	per_level	0.25	blur_resize	0.7 [0.5, 0.9]	0.2 [0.1, 0.3]	0.5 [0.3, 0.7]	1.7
Colour rule	per_level	0.5	area	0.1 [0.0, 0.1]	0.2 [0.1, 0.3]	-0.1 [-0.2, 0.0]	0.6
Colour rule	per_level	0.5	blur_resize	0.1 [0.0, 0.1]	0.1 [0.0, 0.2]	-0.0 [-0.2, 0.1]	0.6

Real Sentinel-2 vs simulated 10 m, Las Vegas (200 chips)

Same chip footprints and labels. Sentinel-2 L2A crops (20 by 20 px), upsampled like the simulation. Oracle F1 is the oracle-location baseline on the real crops.

Detector	Setting	IoU	Simulation	Real S2 F1 [95% CI]	Simulated F1 [95% CI]	Real minus simulated [95% CI]	Oracle F1
OWL-ViT	fixed	0.25	area	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWL-ViT	fixed	0.25	blur_resize	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWL-ViT	fixed	0.5	area	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWL-ViT	fixed	0.5	blur_resize	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWL-ViT	per_level	0.25	area	2.4 [1.7, 3.2]	3.3 [2.5, 4.3]	-0.9 [-1.4, -0.4]	1.7
OWL-ViT	per_level	0.25	blur_resize	2.4 [1.7, 3.2]	2.6 [1.9, 3.6]	-0.2 [-0.8, 0.3]	1.7
OWL-ViT	per_level	0.5	area	0.9 [0.5, 1.2]	1.1 [0.8, 1.5]	-0.3 [-0.5, -0.0]	0.4
OWL-ViT	per_level	0.5	blur_resize	0.9 [0.5, 1.2]	1.0 [0.6, 1.5]	-0.2 [-0.5, 0.1]	0.4
OWLv2	fixed	0.25	area	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWLv2	fixed	0.25	blur_resize	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWLv2	fixed	0.5	area	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWLv2	fixed	0.5	blur_resize	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0
OWLv2	per_level	0.25	area	10.4 [8.8, 12.3]	20.7 [18.8, 22.7]	-10.3 [-11.8, -8.8]	12.1
OWLv2	per_level	0.25	blur_resize	10.4 [8.8, 12.3]	15.0 [12.7, 17.7]	-4.6 [-6.2, -3.1]	12.1
OWLv2	per_level	0.5	area	2.2 [1.6, 2.9]	6.3 [5.4, 7.4]	-4.1 [-5.1, -3.2]	3.1
OWLv2	per_level	0.5	blur_resize	2.2 [1.6, 2.9]	5.2 [4.1, 6.6]	-3.0 [-4.1, -2.0]	3.1
Colour rule	fixed	0.25	area	5.0 [4.0, 6.3]	0.3 [0.0, 0.6]	4.7 [3.7, 5.9]	6.1
Colour rule	fixed	0.25	blur_resize	5.0 [4.0, 6.3]	0.1 [0.0, 0.2]	4.9 [3.9, 6.2]	6.1
Colour rule	fixed	0.5	area	1.6 [1.1, 2.3]	0.0 [0.0, 0.1]	1.6 [1.1, 2.2]	2.2
Colour rule	fixed	0.5	blur_resize	1.6 [1.1, 2.3]	0.0 [0.0, 0.0]	1.6 [1.1, 2.3]	2.2
Colour rule	per_level	0.25	area	4.0 [3.1, 5.1]	0.3 [0.0, 0.7]	3.7 [2.8, 4.8]	5.6
Colour rule	per_level	0.25	blur_resize	4.0 [3.1, 5.1]	0.1 [0.0, 0.2]	3.9 [3.0, 5.0]	5.6
Colour rule	per_level	0.5	area	1.2 [0.9, 1.6]	0.1 [0.0, 0.3]	1.1 [0.8, 1.5]	2.1
Colour rule	per_level	0.5	blur_resize	1.2 [0.9, 1.6]	0.0 [0.0, 0.0]	1.2 [0.9, 1.6]	2.1